

# DERMS:

Distributed Energy Resource Management Systems

# ORCHESTRATING

# THE FUTURE SMART GRID

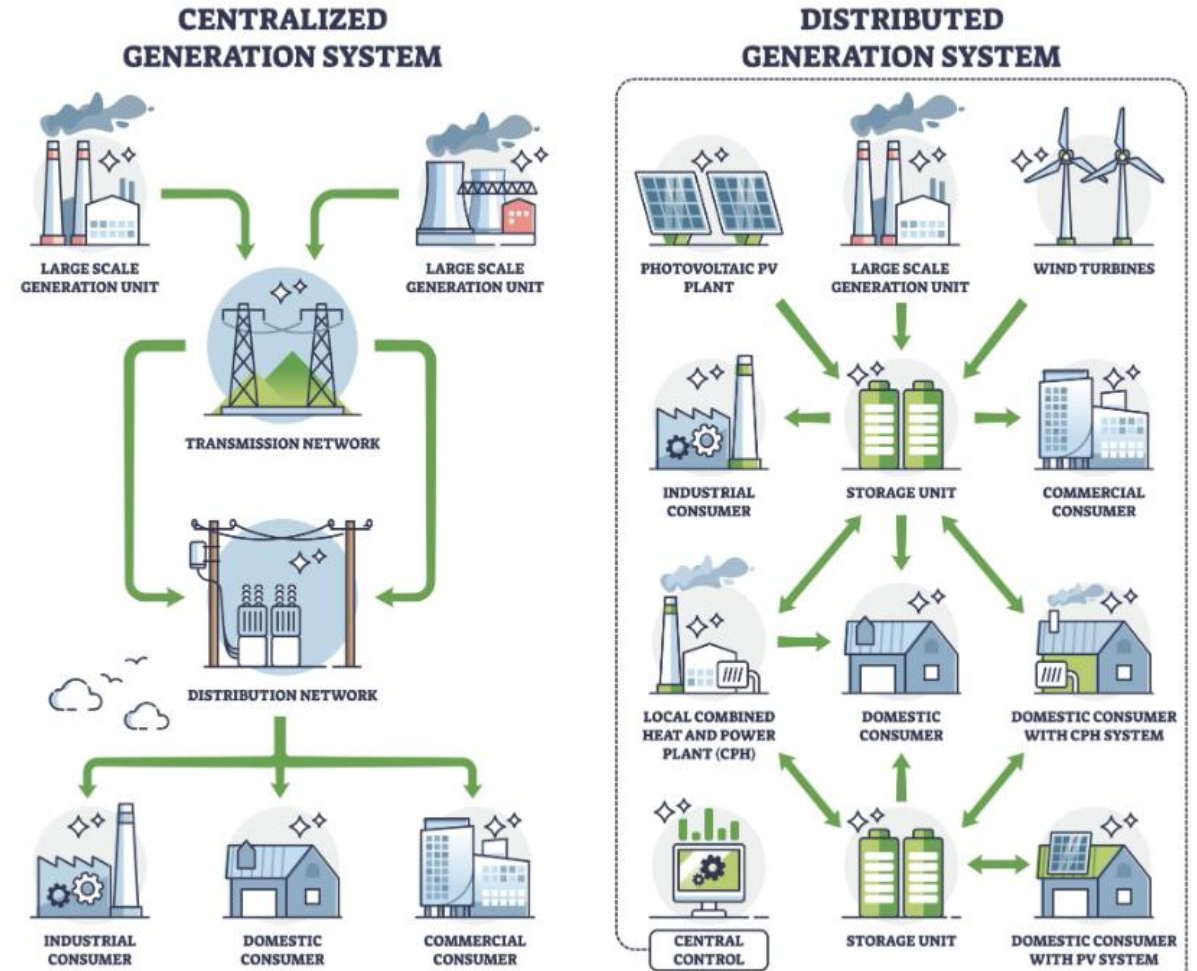
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# The New Energy Landscape

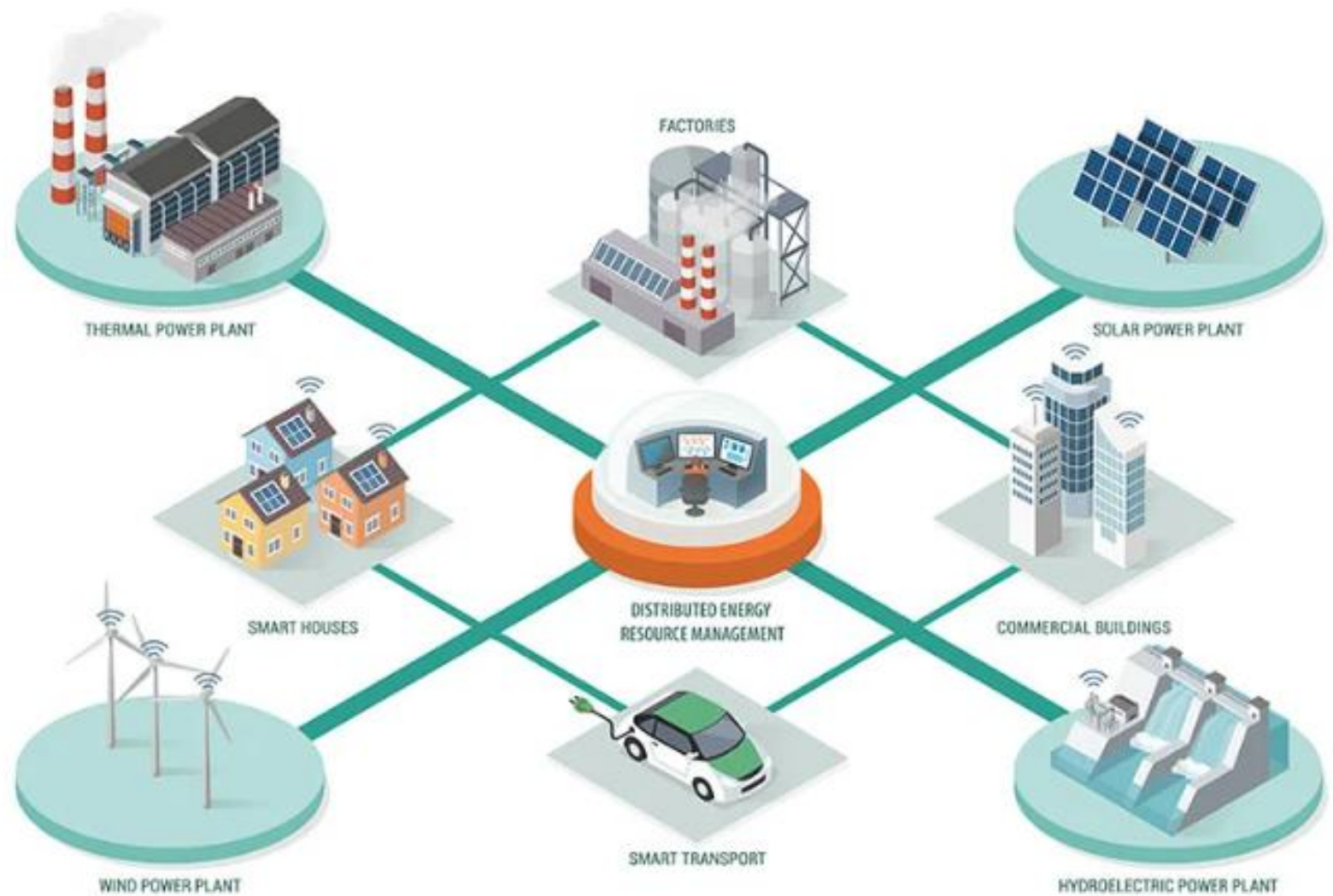
The grid is evolving from a centralized one-way system to a decentralized web of active participants.

## Key Drivers:

- ✓ **Structural:** High-volume and rapid penetration of Distributed Energy Resources (DERs)
- ✓ **Operational:** Managing the Duck Curve
- ✓ **Economic:** Monetization of aggregate power plant resource



# | Enter DERMS



# | DERMS

Distributed Energy Resource Management System (DERMS): A software platform that manages diverse DERs to ensure grid reliability and optimization.



## Visibility

Aggregates data from hundreds to millions of devices to provide a real-time map of what is happening at the grid edge.



## Control

Directly communicates with inverters and controllers to curtail export, ramp up generation, or store energy.



## Optimization

Uses algorithms to balance supply and demand dynamically at the lowest cost and highest reliability.

# | DERMS

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## Forecasting

Predicts generation of renewable energy sources (solar, wind) and load demand using historical data and algorithms.



## Ancillary Services

Coordinates with DERs to provide grid support functions such as frequency and voltage regulation.



## Market Orchestration

Interfaces with electricity markets to participate in Demand Response Programs (DRPs), sell ancillary services, and employ energy arbitrage.

# System Architecture

Consists of three layers:

## 1. Field-level DER layer

- Smart meters, inverters, sensors, storage devices, Electric Vehicle (EV) chargers
- Device identification and registration

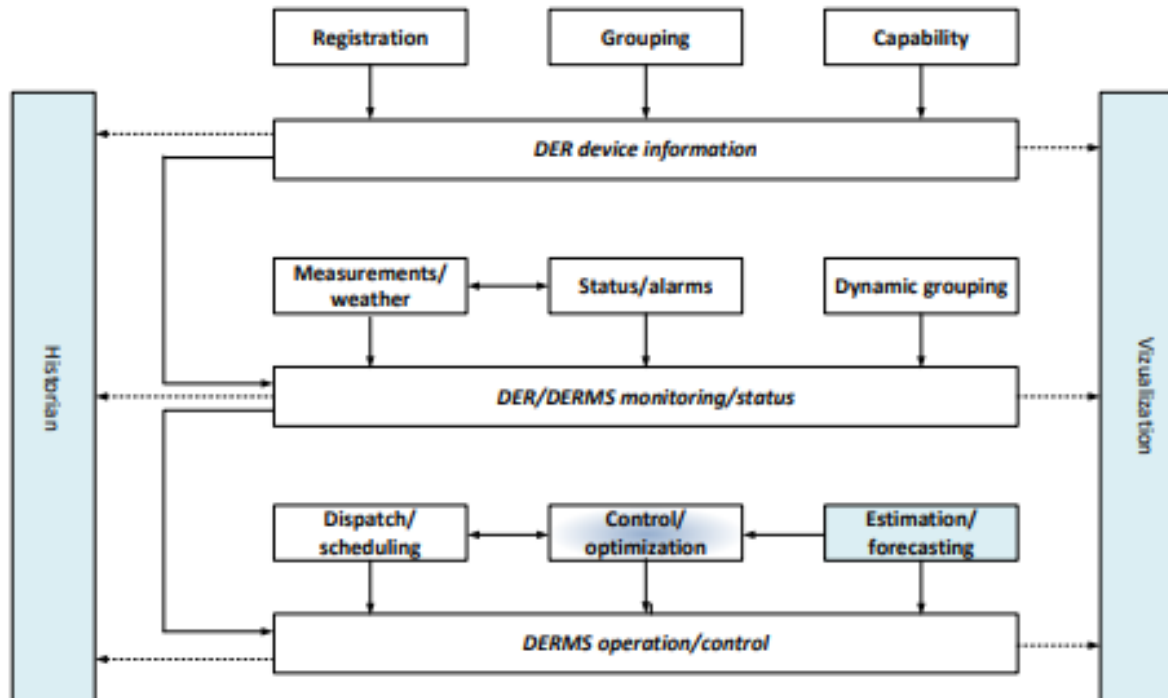
## 2. Communication and Data layer

- Internet of Things (IoT), Advanced Metering Infrastructure (AMI), Supervisory Control and Data Acquisition (SCADA)
- Data collection, monitoring, and transfer

## 3. Control and Optimization layer

- Complex algorithms for core DERMS functions

## DERMS Functional Architecture



# Architecture Patterns

Centralized, federated, third-party

# | DERMS vs. VPPs

## Aggregation is Power

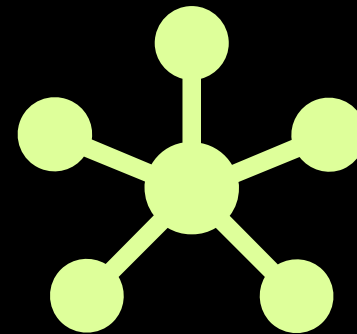
A Virtual Power Plant (VPP) is a network of decentralized DERs controlled through a central system to provide grid services without a dependency on each asset's location.

### Difference

DERMS provides location-specific capabilities of orchestrating at the distribution feeder level.

### Similarity

Both DERMS and VPP aggregate individual DERs into a single controllable asset.



# Benefits

- Enables higher penetration of renewables
- Reduces operational costs
- Improves grid resilience and stability
- Lowers curtailment of renewable energy
- Provides real-time visibility and control of DERs
- Facilitates decarbonization and promotes smart city realization
- Supports cost savings and revenue generation

# Challenges

- Requires robust and secure IoT/AMI communication networks
- Inconsistent location-specific regulatory hurdles
- Scalability challenges pertaining to managing large volume of DERs
- Large upfront cost (millions of dollars)
- Forecasting complexity
- Data management security risks
- Device compatibility challenges

# | Core IEEE Standards for DERMS and DER Integration

## **IEEE 2030.11: DERMS Functional Specification**

Provides functional specification, guiding principles, and a set of core functions for developing and deploying DERMS.

## **IEEE 2030.5: Communication and Interoperability**

A communication standard for DER systems and defines a standardized protocol for communication between utilities, DERMS, and DERs.

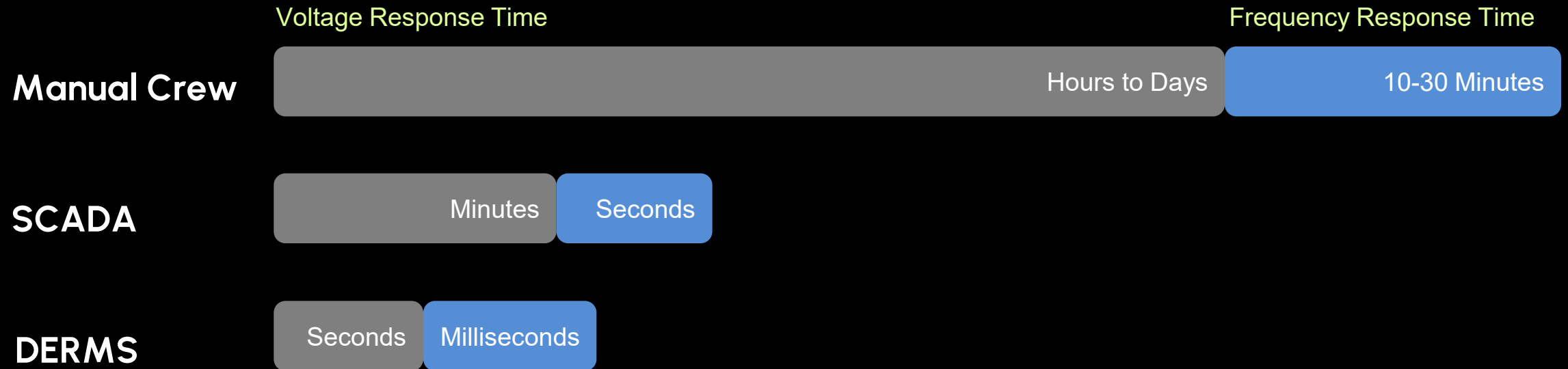


## **IEEE 1547: Interconnection**

Foundational standard that defines criteria and requirements for DERs to safely and reliably connect to the grid.

# | Efficiency Gains: Response Time

Time taken to rectify a localized voltage violation and frequency deviation.



# Canadian Application & Key Facts

## Smart Grid Nova Scotia (SGNS) Project

**Goal:** To understand customer and grid benefits of DER deployments managed by DERMS.

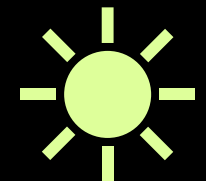
**Set-up:** Installed 134 residential batteries, 3 solar & battery systems, 2 MW community solar garden, 100 EV smart chargers, 2 bi-directional chargers

### Outcomes:

- ✓ EV charging demonstrated large energy and cost savings, and reduction in greenhouse gas emissions
- ✓ Residential, commercial, industrial batteries demonstrated grid smoothing capabilities and emission reductions
- ✓ Smart solar inverters provided low-cost renewable energy and contributed to grid reliability

DERMS received the second largest amount of public smart grids funding between 2018 and 2021 totaling **\$163 million**.

In 2022, renewable sources made up **16.9%** of Canada's total primary energy supply. In 2023, **66%** of total electricity generation in Canada was derived from renewable sources.



# | The Future: AI Integration

## Predictive, Autonomous, Economic Grid Benefits

The next generation of DERMS will rely heavily on Artificial Intelligence (AI) to process the massive influx of IoT data.

- ✓ **Local Forecasting & Optimization:** Predict solar output and DER failures, optimize dispatch scheduling
- ✓ **Self-Healing Grids:** Use Reinforcement Learning (RL) to learn optimal control strategies, decentralize coordination of DERs, detect cyber attacks
- ✓ **Dynamic Pricing:** Real-time automated energy arbitrage and provision of ancillary services



**Thank  
you**

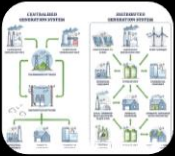
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# | Image Sources



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